Open problems:
Schrödinger equation is equivalent to information conservation

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Classical counterpart.** State and prove the classical analogue: Hamiltonian evolution is equivalent to preservation of the Shannon entropy and of "the definitional independence of degrees of freedom". (Brief, Sec. 4: "This parallels the classical case where Hamiltonian evolution is equivalent to the preservation of Shannon entropy (Liouville's theorem)".)

2. **Entropy conservation as determinism and reversibility.** The brief reads DR-INFO as "an information theoretic way to characterize determinism and reversibility". A precise definition of deterministic and reversible evolution, and a proof that it is equivalent to DR-INFO, are not given. (Brief, Sec. 1.)

3. **Propagator for time-dependent $H(t)$.** "The Schrödinger equation gives rise to a differentiable unitary propagator $U_{t_0 \to t_1}$" is assumed. The formalization proves existence for constant H only; for bounded $H(t)$ it is the linear ODE existence theorem, and for unbounded self-adjoint $H(t)$ it needs Kato-type hypotheses. (Brief, Sec. 3, proof of Theorem 2 ($\Rightarrow$); Log, "Differences from the brief".)

From the formalization

Beyond finite dimension the formalization has a part in any dimension; it leaves open the items "Unbounded Hamiltonians", "Norm-differentiable $U(t)$ in infinite dimension" and "Isometric semigroup without a past" below.

1. **Unbounded Hamiltonians.** Step 3 assumes $U_{t_0 \to t_0 + dt} = I + A(t_0)dt + o(dt)$, and EVDIFF on every density operator forces a bounded H (Banach–Steinhaus). The right form of EVDIFF for unbounded $H(t)$, differentiable only on a dense set of states, and the corresponding version of Theorem 2 are open here. (Log, "Problems found in the brief", Step 3 item and last item of the any-dimension part.)

2. **Mixed states.** $\Phi_{t_0 \to t_1}(\rho) = U_{t_0 \to t_1} \rho\, U_{t_0 \to t_1}^{\dagger}$ for mixed ρ is not formalized. It needs Φ affine (Kadison); whether preservation of equal mixtures alone suffices is not settled. (Log, "Differences from the brief", last item of the finite-dimensional part.)

3. **Equation at every time.** The formalization gives $U(t)$ only for t near t_0 and the Schrödinger equation only at t_0. A global differentiable unitary propagator $U_{t_0 \to t_1}$ for all times, with phases

chosen consistently with $\Phi_{t_1\to t_2}\circ\Phi_{t_0\to t_1}=\Phi_{t_0\to t_2}$, is not built. (Log, "Differences from the brief", item on `schrodinger_of_preserves`.)

4. **Norm-differentiable $U(t)$ in infinite dimension.** In any dimension only the strong equation $\psi'(t_0)=(i\hbar)^{-1}H\psi(t_0)$ is proved; the phase-fixing basis sum does not converge in operator norm. Whether a phase choice makes $t\mapsto U(t)$ differentiable in operator norm is open. (Log, any-dimension part, "Differences from the brief", EVDIFF item.)

5. **Isometric semigroup without a past.** The right shifts V_t on $L^2(0,\infty)$ give an evolution that satisfies [B]EVMIX and preserves the entropy but is induced by no unitary; this example, and the Wold–Cooper reduction of every such semigroup to it, are not formalized. (Log, any-dimension part, "Problems found in the brief", second item.)